

Teorema Fundamental del algebra (Division sintetica)

$$p(x) = x^4 - x^3 + 8x^2 - 7x + 16 \quad x = 2i \quad (1)$$

$$= 2-i \quad (2)$$

$$\begin{array}{r|rrrrr} 1 & -1 & 8 & -7 & 16 \\ & 2i & -2i-7 & 7+8i & -16 \\ \hline 1 & -1+2i & -2i-7 & 8i & 0 \end{array} \quad \begin{array}{l} 2i \\ (x-2i) \end{array}$$

$$x^3 + x^2(-1+2i) + x(-2i-7) + 8i$$

$$\begin{array}{l} 2i(-1+2i) \quad 2i \cdot 8i \\ -2i + 4i^2 \quad 16i^2 \\ -2i - 4 \quad -16 \end{array}$$

$$\begin{array}{r|rrrr} 1 & -1+2i & -2i-7 & 8i \\ & -2i & 2i & -8i \\ \hline 1 & -1 & 7 & 0 \end{array} \quad \begin{array}{l} -2i \\ 2i(-2i-7) \\ -4i^2 - 14i \\ 4 + 8i \end{array} \quad \begin{array}{l} (x-(-2i)) \\ (x+2i) \end{array}$$

$$(x^2 - x + 7)(x-2i)(x+2i)$$

$$\begin{array}{ccc} x^2 & -x & +7 \\ a & b & c \end{array}$$

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (-1)^2 - 4 \cdot 1 \cdot 7 \\ &= -15 \end{aligned}$$

$$\frac{-1 \pm i\sqrt{15}}{2}$$

$$(x-2)(x+2i) \left(x - \frac{-1+i\sqrt{15}}{2} \right) \left(x - \frac{-1-i\sqrt{15}}{2} \right)$$

$$P(x) = 7x^3 - x - 7ix^2 + i \quad x = i \text{ (1)}$$

$$= 7x^3 - 7ix^2 - x + i \quad x = -i \text{ (2)}$$

$$\begin{array}{r|l} 7 & -7i & -1 & i \\ & 7i & 0 & -i \\ \hline 7 & 0 & -1 & 0 \end{array} \quad (x-i)$$

$$7x^2 + 0x - 1$$

$$a^2 - b^2 = 7x^2 - 1^2$$

$$(2x)^2 - 1^2$$

$$7x^2 - 1 = (2x-1)(2x+1)$$

$$(x-i)(2x-1)(2x+1)$$

$$\boxed{x = i \quad x = \frac{1}{2} \quad x = -\frac{1}{2}}$$

Parces Complejas $x^n = z$

Recordar

$$z = a + bi \rightarrow \text{Rectangular}$$

$$z = r(\cos(\theta) + i\sin(\theta)) \rightarrow \text{Polar}$$

$$\begin{aligned} & \begin{cases} \cos(\theta) + i\sin(\theta) \rightarrow \arctan\left(\frac{b}{a}\right) \\ \sqrt{a^2 + b^2} \end{cases} \end{aligned}$$

Dependiendo de esta tabla lo de $\tan^{-1}\left(\frac{b}{a}\right)$ hay sumar o restar π

Cuadrante	a	b	Rango	Que hago?
I	+	+	Ya esta	NADA
II	-	+	Esta fuera	$+\pi$ al θ
III	-	-	Esta fuera	$-\pi$ al θ
IV	+	-	Ya esta	NADA

$$\theta = \frac{\pi}{2} \quad \text{si } a = 0 \quad \text{y } b > 0 \quad z = 3i$$

$$\theta = -\frac{\pi}{2} \quad \text{si } a = 0 \quad \text{y } b < 0 \quad z = -3i$$

$$\theta = \left[\tan^{-1}\left(\frac{b}{a}\right) \right] \pm \pi$$

1) Factorizar

2) Escribir en forma polar

3) Usar teorema de de Moivre

Tener en cuenta

$$z = r(\cos(\theta) + i\sin(\theta))$$

Hay que probar

$n-1$ soluciones

$$x^n = r(\cos(\theta) + i\sin(\theta))$$

$$n = k, \quad k = 0, 1, \dots, n-1$$

$$x = \sqrt[n]{r} \cdot \left(\cos\left(\frac{\theta + 2\pi k}{n}\right) + i\sin\left(\frac{\theta + 2\pi k}{n}\right) \right)$$

Ejemplo $x^n = z$

$$x^3 - 8i = 0$$

b a

$$x^3 = 8i + 0$$

$$r = \sqrt{8^2 + 0^2} = 8$$

$$\theta = \alpha = 0 \wedge \text{b} > 0 = \frac{\pi}{2}$$

$$x^3 = 8 \cdot \text{cis}\left(\frac{\pi}{2}\right)$$

$$\sqrt[3]{8} \cdot \text{cis}\left(\frac{\frac{\pi}{2} + 2\pi \cdot k}{3}\right)$$

= 2

$$n=3, \quad k=0, 1, 2 \quad (n-1)$$

$$k=0 \quad 2 \cdot \text{cis}\left(\frac{\frac{\pi}{2} + 2\pi \cdot 0}{3}\right)$$

$$x_1 = 2 \cdot \text{cis}\left(\frac{\pi}{6}\right)$$

$$k=1 \quad 2 \cdot \text{cis}\left(\frac{\frac{\pi}{2} + 2\pi \cdot 1}{3}\right)$$

$$x_2 = 2 \cdot \text{cis}\left(\frac{5\pi}{6}\right)$$

$$k=2 \quad 2 \cdot \text{cis}\left(\frac{\frac{\pi}{2} + 2\pi \cdot 2}{3}\right)$$

$$x_3 = 2 \cdot \text{cis}\left(\frac{9\pi}{6}\right)$$

$$x^6 + 1 = 0 \quad r = \sqrt{(-1)^2 + 0^2} = 1$$

$$x^6 = -1 + 0i \quad \theta = \tan^{-1}\left(\frac{0}{-1}\right) + \pi = \pi$$

$$x^6 = 1 \operatorname{cis}(\pi) \quad k = 0, 1, 2, 3, 4, 5 \quad (n-1)$$

$$= \sqrt[6]{1} \cdot \operatorname{cis}\left(\frac{\pi + 2\pi k}{6}\right)$$

$$k=0 \rightarrow x_1 = \operatorname{cis}\left(\frac{\pi}{6}\right)$$

$$k=1 \rightarrow x_2 = \operatorname{cis}\left(\frac{\pi}{2}\right)$$

$$k=2 \rightarrow x_3 = \operatorname{cis}\left(\frac{5\pi}{6}\right)$$

$$k=3 \rightarrow x_4 = \operatorname{cis}\left(\frac{7\pi}{6}\right)$$

$$k=4 \rightarrow x_5 = \operatorname{cis}\left(\frac{3\pi}{2}\right)$$

$$k=5 \rightarrow x_6 = \operatorname{cis}\left(\frac{11\pi}{6}\right)$$

Magnitudes

$$|w| = |z \cdot x| = |z| \cdot |x|$$

Siempre en polar $\text{Arg}(w) = \theta$
 $|w| = r = \sqrt{a^2 + b^2}$

$$|\overline{w}| = |w|$$

3) Determine la forma polar de todos los números complejos que cumplen, simultáneamente, las condiciones que se muestran en cada caso:

a)
$$\begin{cases} |w| = |(3-i) \cdot 2i - 1| \\ \text{Arg}(w) = \text{Arg}\left[\frac{(-1+i)^5}{\sqrt{3}-i}\right] \end{cases} \quad |\overline{w}| = |w|$$

$$|w| = |(3-i)(2i-1)|$$

$$\begin{aligned} r_1 &= \sqrt{3^2 + (-1)^2} & r_2 &= \sqrt{2^2 + (-1)^2} \\ &= \sqrt{10} & &= \sqrt{5} \end{aligned}$$

$$|w| = 5\sqrt{2}$$

$$\text{Arg}(w) = \text{Arg}\left(\frac{\overbrace{(-1+i)^5}^{z_1}}{\underbrace{\sqrt{3}-i}_{z_2}}\right)$$

$$\begin{aligned} z_1 &= -1 + i & z_2 &= \sqrt{3} - i \\ \theta &= \arctan\left(\frac{1}{-1}\right) + \pi & \theta &= \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) \\ &= \frac{3\pi}{4} & &= -\frac{\pi}{6} \\ &\text{cis}\left(\frac{3\pi}{4}\right) & &\text{cis}\left(-\frac{\pi}{6}\right) \end{aligned}$$

$$\begin{aligned} \text{Arg}\left(\frac{\overbrace{(-1+i)^5}^{z_1}}{\underbrace{\sqrt{3}-i}_{z_2}}\right) &\rightarrow \frac{\text{cis}\left(\frac{3\pi \cdot 5}{4}\right)}{\text{cis}\left(-\frac{\pi}{6}\right)} \rightarrow \frac{\text{cis}\left(\frac{15\pi}{4} - -\frac{\pi}{6}\right)}{\text{cis}\left(\frac{97\pi}{12}\right)} \\ &= \frac{[\text{cis}\left(\frac{3\pi}{4}\right)]^5}{\text{cis}\left(-\frac{\pi}{6}\right)} = \frac{\text{cis}\left(\frac{15\pi}{4}\right)}{\text{cis}\left(-\frac{\pi}{6}\right)} \rightarrow \boxed{w = 5\sqrt{2} \cdot \text{cis}\left(\frac{97\pi}{12} + 2\pi k\right)} \end{aligned}$$

De radianes a grados
radianes, $\frac{180}{\pi}$

Cuadrante	a	b
1) $0 < \theta < \frac{\pi}{2}$ $0 < a < 90$	> 0	> 0
2) $\frac{\pi}{2} < \theta < \pi$ $90 < a < 180$	< 0	> 0
3) $\pi < \theta < \frac{3\pi}{2}$ $180 < a < 270$	< 0	< 0
4) $\frac{3\pi}{2} < \theta < 2\pi$ $270 < a < 360$	> 0	< 0

En cualquier caso que el arg NO
sea $\pm \pi$ v $\pm \frac{\pi}{2}$ para sacar el
ángulo se $\theta = \left(\frac{b}{a} = \tan(\theta) \right)$

$$d) \begin{cases} |w - 3i + 3| = \text{Im}(\overline{7 - 5i}) \\ \text{Arg}(w + 3i) = 3\pi/4 \end{cases} \quad z, w = a + bi$$

$$\text{Arg}(w + 3i) = \frac{3\pi}{4}$$

$$a + bi + 3i = \frac{3\pi}{4}$$

$$a + (b+3)i = \frac{3\pi}{4}$$

$$\frac{b+3}{a} = \tan\left(\frac{3\pi}{4}\right) \quad \frac{3\pi}{4}, \frac{180}{\pi}$$

$$= 135$$

$$\frac{b+3}{a} = -1 \quad \leftarrow \text{Calcu}$$

$$2 \quad 90 < 135 < 180$$

$$a < 0 \quad b > 0$$

$$b+3 = -a$$

$$\begin{cases} b = -a - 3 \\ a = -b - 3 \end{cases} \quad \begin{matrix} 1 \\ 2 \end{matrix} \quad \begin{matrix} \text{Resolver magnitud} \\ \text{para continuar} \end{matrix}$$

$$|w - 3i + 3| = \text{Im}(\overline{7 - 5i}) \quad \overline{7 - 5i} = 7 + 5i$$

$$|a + bi - 3i + 3| = 5 \quad = 5$$

$$|(a+3) + (b-3)i| = 5 \quad \text{Todos al}$$

↙ Cuadrado

$$(a+3)^2 + (b-3)^2 = 5^2$$

$$\begin{aligned} (a+3)^2 + (b-3)^2 &= 5^2 & b &= -a-3 \\ (-b-3+3)^2 + (b-3)^2 &= 25 & a &= -b-3 \end{aligned} \quad \left. \vphantom{\begin{aligned} (a+3)^2 + (b-3)^2 &= 5^2 \\ (-b-3+3)^2 + (b-3)^2 &= 25 \end{aligned}} \right\} \begin{matrix} 1 \\ 2 \end{matrix}$$

$$b^2 + b^2 - 6b + 9 = 25$$

$$2b^2 - 6b - 16 = 0$$

$$b_1 = \frac{3 + \sqrt{91}}{2} \quad b_2 = \frac{3 - \sqrt{91}}{2}$$

$$\approx 9,7$$

$$\approx -1,7$$

$$a = -b-3 \quad \begin{cases} b = 9,7 \rightarrow -9,7-3 = -12,7 \\ b = -1,7 \rightarrow -1,7-3 = -4,7 \end{cases}$$

$$\boxed{11 \quad -12,7 + 9,7 ;}$$

$$e) \begin{cases} |z-2| = 5 \\ \text{Arg}(z-1) = 3\pi/4 \end{cases}$$

$$|a+bi-2| = 5$$

$$|(a-2) + (b)i| = 5$$

$$(a-2)^2 + b^2 = 25 \quad \rightarrow \text{Can't solve yet}$$

$$\text{Arg}(z-1) = \frac{3\pi}{4}$$

$$a+bi-1 = \frac{3\pi}{4}$$

$$(a-1) + (b)i = \frac{3\pi}{4}$$

$$\frac{b}{a-1} = \tan\left(\frac{3\pi}{4}\right)$$

$$\frac{3\pi}{4}, \frac{180}{\pi} = 135$$

$$90 < 135 < 180$$

2do Cuadrante

$$a < 0 \quad b > 0$$

$$\frac{b}{a-1} = -1$$

$$b = 1-a$$

$$a = 1-b$$

$$a < 0$$

$$b > 0$$

$$(a-2)^2 + b^2 = 25$$

$$(1-b-2)^2 + b^2 = 25$$

$$(-b-1)^2 + b^2 = 25$$

$$b^2 + 2b + 1 + b^2 = 25$$

$$2b^2 + 2b - 24 = 0$$

$$2b^2 + 2b - 27 = 0$$

B.3

$$b = 1 - a$$

$$2b \quad -6 = -6b$$

3-8

$$a = 1 - b$$

$$b \quad 7 = \frac{8b}{2b}$$

12.7

$$(2b - 6)(b + 7)$$

$$2b - 6 = 0$$

$$b + 7 = 0$$

$$a < 0$$

$$b > 0$$

$$b = 3$$

$$b = -7$$

$$b = 3$$

$$b = 1 - a$$

$$a = 1 - b$$

$$a = 1 - 3 = -2$$

$$\boxed{-2 + 3i}$$